Machine Learning for Physicists - Homework 2 - Part II: Representing a structured function with a vision transformer

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Question a: Parameter Counting

How many parameters does the transformer have?

We analyze the components of the simplified Vision Transformer defined in the exercise:

- Model dimension: $d = 4$
- Number of heads: $h = 2$
- Grid size: $T = s^2/4$

1. Parameters of matrices $V^{(h)}$ (Values)
Each head has a value matrix $V^{(h)} \in \mathbb{R}^{d/2 \times d} = \mathbb{R}^{2 \times 4}$.

$$2 \text{ heads} \times (2 \times 4) \text{ weights} = 16 \text{ parameters}$$

2. Attention parameters $\alpha_{ij}^{(h)}$ (Factored Attention)
The problem states that α_{ij} are learnable parameters depending only on position. Since there are no queries/keys, we learn the attention map directly.

$$2 \text{ heads} \times (T \times T) \text{ weights} = 2T^2 \text{ parameters}$$

3. MLP parameters (W and c)
The MLP is defined as $z = \log\cos(W\overline{x}_i + c)$ with $W \in \mathbb{R}^{d \times d}$ and $c \in \mathbb{R}^d$.

- Matrix W: $4 \times 4 = 16$ parameters.
- Bias vector c: Since $z \in \mathbb{R}^d$, the bias must be a vector of size d. It cannot be a scalar, otherwise we could not selectively bias some dimensions while leaving others unbiased (needed for the zero-padding trick). Thus, c contributes 4 parameters.

Total MLP: $16 + 4 = 20$ parameters.

4. Total number of parameters Summing the contributions from the value matrices, attention maps, and the MLP, we obtain:

$$N_{\text{total}} = 16 + 2T^2 + 20 = \boxed{2T^2 + 36}$$

Remark: The parameter count scales quadratically with the sequence length T (i.e., $N_{total} \sim \mathcal{O}(T^2)$). For large T, the attention map weights dominate, rendering the constant terms negligible.

Question b: Model Weights

Determine the values of $V^{(h)}, \alpha_{ij}^{(h)}, W$, and c for which $f_{\mathbf{trans}}(\sigma) = f(\sigma)$. The target function is:

$$f(\sigma) = \sum_{i=1}^{T} \left[\log\cos\left(\frac{\pi}{2} + \frac{\pi}{4}(\sigma_{i,4} + 3\sigma_{P(i),1})\right) + \log\cos\left(\frac{\pi}{2} + \frac{\pi}{4}(3\sigma_{i,2} + \sigma_{i,3})\right)\right] \tag{1}$$

Our transformer output is $f_{\text{transf}}(\sigma) = \sum_{i=1}^{T}\sum_{\mu=1}^{d} z_{i,\mu}$. To match the target, we assign specific roles to the dimensions of the embedding vector.

1. Value Matrices $V^{(h)}$

We define $x_i = (\sigma_{i,1}, \sigma_{i,2}, \sigma_{i,3}, \sigma_{i,4})^\top$. We use the hint structure, correcting the coefficients to match the target $(3\sigma_{i,2} + \sigma_{i,3})$:

$$V^{(1)} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 1 & 0 \end{pmatrix} \quad \Rightarrow \quad v_i^{(1)} = \begin{pmatrix} \sigma_{i,4} \\ 3\sigma_{i,2} + \sigma_{i,3} \end{pmatrix} \tag{2}$$

$$V^{(2)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \Rightarrow \quad v_i^{(2)} = \begin{pmatrix} \sigma_{i,1} \\ 0 \end{pmatrix} \tag{3}$$

2. Factored Attention α_{ij}

Head 1 looks at the patch itself (Self), Head 2 looks at the neighbor $P(i)$ (Neighbor).

- **Head 1 (Identity):** $\alpha_{ij}^{(1)} = 1$ if $j = i$, else 0.
- **Head 2 (Permutation):** We need to fetch $\sigma_{P(i),1}$ to position i. Thus, we attend to j such that the value comes from $P(i)$.

$$\alpha_{ij}^{(2)} = \begin{cases} 1 & \text{if } j = P(i) \\ 0 & \text{otherwise} \end{cases}$$

The concatenated context vector $\overline{x}_i \in \mathbb{R}^4$ becomes:

$$\overline{x}_i = \text{Concat}(v_i^{(1)}, A_i^{(2)}) = \begin{pmatrix} \sigma_{i,4} \\ 3\sigma_{i,2} + \sigma_{i,3} \\ \sigma_{P(i),1} \\ 0 \end{pmatrix} \tag{4}$$

3. MLP Weights (W, c)

We need to combine these components to form the arguments of the $\log\cos$. Target arguments:

1. Term 1: $\frac{\pi}{4}(\sigma_{i,4} + 3\sigma_{P(i),1}) + \frac{\pi}{2}$
2. Term 2: $\frac{\pi}{4}(3\sigma_{i,2} + \sigma_{i,3}) + \frac{\pi}{2}$

We choose W to perform the linear mixing and c to add the phase shift:

$$W = \frac{\pi}{4}\begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad c = \begin{pmatrix} \pi/2 \\ \pi/2 \\ 0 \\ 0 \end{pmatrix} \tag{5}$$

4. Verification

We verify that the transformer output matches the target function exactly. First, we compute the affine transformation $\mathbf{a}_i = W\overline{x}_i + c$ for a single patch i.

$$\mathbf{a}_i = \underbrace{\frac{\pi}{4}\begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} \sigma_{i,4} \\ 3\sigma_{i,2} + \sigma_{i,3} \\ \sigma_{P(i),1} \\ 0 \end{pmatrix}}_{\text{Mixing input features}} + \underbrace{\begin{pmatrix} \pi/2 \\ \pi/2 \\ 0 \\ 0 \end{pmatrix}}_{\text{Phase shifts}} = \begin{pmatrix} \frac{\pi}{4}(\sigma_{i,4} + 3\sigma_{P(i),1}) + \frac{\pi}{2} \\ \frac{\pi}{4}(3\sigma_{i,2} + \sigma_{i,3}) + \frac{\pi}{2} \\ 0 \\ 0 \end{pmatrix} \tag{6}$$

Next, the MLP applies the element-wise non-linearity $z_{i,\mu} = \log\cos((\mathbf{a}_i)_\mu)$. We sum these components over the model dimension $\mu = 1 \dots 4$ to get the contribution of patch i:

$$\begin{aligned}
\text{Patch}_i &= \sum_{\mu=1}^{4} z_{i,\mu} \\
&= \log\cos\left(\mathbf{a}_{i,1}\right) + \log\cos\left(\mathbf{a}_{i,2}\right) + \log\cos(0) + \log\cos(0) \\
&= \log\cos\left(\frac{\pi}{2} + \frac{\pi}{4}(\sigma_{i,4} + 3\sigma_{P(i),1})\right) + \log\cos\left(\frac{\pi}{2} + \frac{\pi}{4}(3\sigma_{i,2} + \sigma_{i,3})\right) + \underbrace{0+0}_{\text{since } \cos(0)=1}
\end{aligned}$$

Finally, the total output of the transformer is the sum over all patches $i = 1 \dots T$:

$$f_{\text{transf}}(\sigma) = \sum_{i=1}^{T} \text{Patch}_i \equiv f(\sigma) \tag{7}$$

The construction reproduces the target function $f(\sigma)$ exactly.

5. Results

Summary of the parameters found that solve the problem exactly:

$$\boxed{\begin{array}{ll}
V^{(1)} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 3 & 1 & 0 \end{pmatrix}, & \alpha_{ij}^{(1)} = \begin{cases} 1 & \text{if } j = i \\ 0 & \text{else} \end{cases} \\[2em]
V^{(2)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & \alpha_{ij}^{(2)} = \begin{cases} 1 & \text{if } j = P(i) \\ 0 & \text{else} \end{cases} \\[2em]
W = \frac{\pi}{4}\begin{pmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & c = \frac{\pi}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}
\end{array}} \tag{8}$$

Remark: The proposed solution is not the only one for the problem presented.